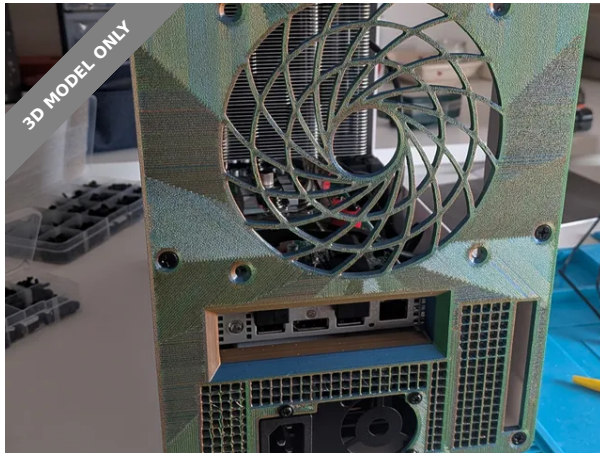




BC-250 Quiet Case - Tower cooler

S Gadget

[VIEW IN BROWSER](#)

updated 12. 7. 2026 | published 12. 7. 2026

Summary

A quiet BC-250 case made to accomodate a tower cooler with an optional digital display, a front USB hub, and an SSD slot

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Tags: [noctua](#) [deepcool](#) [bc250](#)

Description

If you're looking for a **quiet** and relatively easy case for your BC-250, look no further. While not as compact as other designs, this case can accomodate a DeepCool tower cooler (AK400 or AK500 series) and its digital display (optional).

The result is not only whisper quiet operation at idle, but also very low sound level at full load, even with **heavy overclocking**.

Updates

29-mar-2026 : fixed fan rubbing on front panel, ATX breakout board mouting holes, and reinforced mouting guides.

12-jul-2026 : posted files for the generic version (XL) that was used for the Aftermath article, this version loses its top window and accomodates

bigger tower coolers than the AK500.

<https://aftermath.site/bc250-steam-machine-budget-computer/>

Components

- **Flex-ATX PSU**
 - My pick : MetalFish 500W
- **120mm Deepcool Tower Cooler** (AK400/AK500) and **brackets**, from Stanislav Slow
 - Optional : Digital version
 - My pick: AK500 Digital SE
- **140mm front fan** (intake)
- **USB hub** : https://www.amazon.fr/dp/B0D97V3N3R?ref=ppx_yo2ov_dt_b_fed_asin_title
- **ATX breakout board** : https://www.amazon.fr/dp/B0834RNJX5?ref=ppx_yo2ov_dt_b_fed_asin_title
- **16mm latching power button**
 - My pick: https://www.amazon.fr/dp/B07QP6WY7R?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

Recommendations for cool & quiet operation

- **80 x 15mm** fan, over the PSU (replaces the PSU's 40mm fan)
- **120 x 25mm** CPU fan (replaces DeepCool's fan which exhibits some noticable humming)
- And very optionnaly, a **120 x 25 mm** exhaust fan

The fan brands I recommend are **Noctua**, **Arctic** and **Be Quiet**. I have nothing but great experience with those 3. Just pick the one you prefer to work with.

Additional hardware you might want during assembly

- 180° "up" USB adapter(s)
 - 1 to the front USB hub
 - 1 (optional) to the cooler's digital display
- 90° "down" PCIe adapter
- USB-to-motherboard adapter (to connect the Digital Edition display)
- 90° ATX cable adapter
- Y-fan-cables (not needed with Arctic fan, though)

Mouting Instructions

Install M3 brass inserts on the case (all inserts should be 5mm diameter, the dimension specified below is for their length) :

Front case part :

- 2 x 2 x (4mm insert) on both side panels
- 4 x (4mm insert) on front/top side, to mount the USB hub
- 4 x (4mm insert) on front/bottom side, to mount the ATX breakout board
- 2 x (6mm insert) on front/middle support, to mount the bc250 board

Back case part :

- 2 x 2 x (4mm insert) on both side panels
- 4 x (6mm insert) on the front side, these will be used to assemble the front part
- 1 x (6mm insert) on the back side, to mount the bc250 board

You can glue the front and back case parts together with cyanolite, using 4 internal mounting screws to compress the 2 parts after gluing.

Once the case is assembled, install the internal 80mm PSU fan (should blow downside). Remove the PSU's top cover, and slide the PSU in. Take care to route the fan's cable to the front of the case.

Remove the board's front bracket, and connect the ATX 8 pins cable (you won't be able to do it once the board is in place, a 90° connector will greatly help). Slide the board in (backplate in between brackets). Secure the board using 3 x M3 screws, 1 short at the back and 2 long at the front.

Then install the ATX breakout board, and the board support (secure it using a fan screw). Route the USB power supply cable to the ATX breakout board (+12V/GND) using right hand side bridges, and the latching button using left-hand side bridges.

Route USB cable from the hub to the back of the board.

Mount front fan to the front pannel, connect all fans to the FAN1 port, and mount front panel.

Mount back panel, pack your cables on the left case side (here's a bit of room there), and mount side panels.

Plug the PSU to the wall, power-on, enjoy !

Script to enable the digital display on Bazzite OS

Download deepcool-digital-linux to /var/local/bin
<https://github.com/Nortank12/deepcool-digital-linux>

Then :

```
sudo chmod +x /var/local/bin/deepcool-digital-linux
./deepcool-digital-linux
```

Digital display should start showing the CPU temperature.

Time to make it run as a service. First, give system permissions :

```
sudo semanage fcontext -a -t bin_t "/var/local/bin/deepcool-digital-linux"
sudo restorecon -v /var/local/bin/deepcool-digital-linux
```

Then, paste the following in /etc/system/deepcool-digital.service

```
[Unit]
Description=DeepCool Digital
[Service]
ExecStart=/var/local/bin/deepcool-digital-linux
Restart=on-failure
RestartSec=5s
User=[your user name]

[Install]
WantedBy=multi-user.target
```

Lastly, start the service

```
sudo systemctl start deepcool-digital
sudo systemctl status deepcool-digital
```

Thanks

DeepCool AG/AK brackets from Stanislav Slow has been the starting point of this design:

I remixed **Noctua's High Efficiency Fan Grill** to design the case's front and back panels:

Last but not least, **DeepCool Digital Linux service** provided by: <https://github.com/Nortank12/deepcool-digital-linux>

Model files



Deepcool Edition - Digital Display

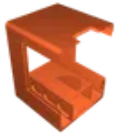
8 files



quiet-bc-case.stl



front-case.stl



back-case.stl



back-panel.stl



front-panel.stl



side-panel.stl



stand.stl

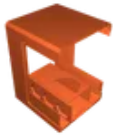


card-support.stl



XL Version - Aftermath

8 files



case-back.stl



panel-back.stl



case-front.stl



panel-front.stl



panel-side.stl



board-support.stl



stand-model-1.stl



stand-model-2.stl

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